Samuel Ogulluk

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EDUCATION

- **École Normale Supérieure Paris-Saclay, Paris** — **2025-Present**

  MSc in Electronic and Digital Engineering

  Department of Electronic and Digital Engineering - Student civil servant
  The key courses included: Microprocessors architecture, Computer science, Signal & Image Processing, Physics, AI.

- **Sorbonne Université, Paris** — **2025-Present**

  BSc in Theoretical Physics - The key courses included: Quantum Physics, Complex Analysis, Material mechanics.

- **École Normale Supérieure de Rennes, Rennes** — **2024-2025**

  BSc in Mechanical Engineering *with the highest honours*

  BSc in Electrical Engineering *with high honours*

  Department of Mechatronics - Student civil servant
  The key courses included : Mechanics, Rigid-Body Mechanics, Engineering Drawing, CAD, Analog Electronics.

- **Lycée Hoche, Versailles** — **2022-2024**

  Classe préparatoire aux grandes écoles d'ingénieurs
  Preparatory class for France's leading engineering Universities -
  Intensive generalist curriculum in Mathematics and Chemistry with a Major in Physics and Engineering

WORK EXPERIENCE

- **CERN, Meyrin, Switzerland — Internship** — **June–July 2025**

  Worked on the FCC (Future Circular Collider) project in the BEAMS department under the supervision of the project engineer Jean-Paul Burnet.

  Developed and benchmarked state-of-the-art robotic pose estimation algorithms, assembled a dedicated testbed, and ensured robustness under critical operating conditions. Attended Summer Student lectures on Particle Physics, Experimental Physics and Computer Sciences.

- **Development of a C Framework** — **2025–Present**

  Creator of **Otternet**, a C framework (Python API in progress) for Machine Learning and Data Science. Provides neural networks, training routines, and modular tools for easy and efficient computation.

LANGUAGES & TECHNICAL SKILLS

I am a native **French** speaker, fully proficient in **English** (C1 IELTS), and I have basic knowledge of **Spanish**

- **electronic board design** (KiCad, PSpice)
- **CAD** (SolidWorks, Autodesk 360)
- **MATLAB**
- 3D printing, laser engraving
- **machine learning**, **computer vision**
- serial and parallel **robotics**
- **Python** (NumPy, JAX ,TensorFlow, OpenCV, Keras)
- **C, C++, and C#**
- LaTeX
- **ARM**assembly
- **web development** (HTML, CSS, Typescript)
- **Bash and Linux** environments

PERSONAL INTERESTS

- **Music** (10+ years: classical/electric guitar, piano, ENS rock band)
- **Sports** (Cycling, Judo - brown belt)
- **Aeronautics & Space** (Aeronautical Training Certificate - full marks)
- Strong interest in **science & technology**